

YEAR 7 • UNIT 2 • EXTRA HOMEWORK

Split, Multiply, Add

Two-digit multiplication • dividing • and the same grid with a letter in it

Name: _____

Class: _____

Date given: _____

Date due: **Monday 21 September 2026****Why this sheet exists**

Nobody multiplies by 16. They **split it up**: 4×10 is 40, 4×6 is 24, and $40 + 24 = 64$.

That is three steps — **split, multiply, add** — and it is also, exactly, what expanding a bracket is. 4×16 and $4(x + 3)$ are drawn in the **same grid**. Part 1 of this sheet practises the grid on numbers. Part 4 puts a letter in one box. Nothing else changes.

No calculator. Write in every box.

Part 1 • Split it up

F1 Tens and ones

Split each number. The first one is done for you.

(a) $47 = 40 + 7$

(d) $18 = \underline{\hspace{2cm}}$

(g) $34 = \underline{\hspace{2cm}}$

(b) $23 = \underline{\hspace{2cm}}$

(e) $52 = \underline{\hspace{2cm}}$

(h) $70 = \underline{\hspace{2cm}}$

(c) $65 = \underline{\hspace{2cm}}$

(f) $96 = \underline{\hspace{2cm}}$

(i) $89 = \underline{\hspace{2cm}}$

F2 Times a ten

This is the box that people get wrong, so do a page of them first.

Multiply, then put the zero back on

$4 \times 30: 4 \times 3 = 12, \text{ so } 4 \times 30 = 120.$

$20 \times 30: 2 \times 3 = 6, \text{ two zeros, so } 20 \times 30 = 600.$

(a) $4 \times 30 = \underline{\hspace{2cm}}$

(d) $9 \times 60 = \underline{\hspace{2cm}}$

(g) $20 \times 30 = \underline{\hspace{2cm}}$

(j) $30 \times 80 = \underline{\hspace{2cm}}$

(b) $6 \times 50 = \underline{\hspace{2cm}}$

(e) $8 \times 20 = \underline{\hspace{2cm}}$

(h) $40 \times 50 = \underline{\hspace{2cm}}$

(k) $50 \times 50 = \underline{\hspace{2cm}}$

(c) $7 \times 40 = \underline{\hspace{2cm}}$

(f) $3 \times 90 = \underline{\hspace{2cm}}$

(i) $60 \times 70 = \underline{\hspace{2cm}}$

(l) $20 \times 90 = \underline{\hspace{2cm}}$

F3 One row: two digits by one digit

Fill the two boxes, then add them.

The grid, once, in full 4×16 means $4 \times (10 + 6)$. One box, one multiplication.

$\times$	10	6
4	40	24

$40 + 24 = 64$

Split 16 into $10 + 6$. **Multiply** each box. **Add** the boxes.

$\times$	20	3
7		

$$\text{(a) } 7 \times 23$$

$$= \underline{\hspace{2cm}}$$

$\times$	40	8
6		

$$\text{(b) } 6 \times 48$$

$$= \underline{\hspace{2cm}}$$

$\times$	50	4
8		

$$\text{(c) } 8 \times 54$$

$$= \underline{\hspace{2cm}}$$

$\times$	30	7
9		

$$\text{(d) } 9 \times 37$$

$$= \underline{\hspace{2cm}}$$

F4 Now without the printed grid

Draw your own grid in the box, then write the answer under it.

6×34	8×72	7×85	9×46
5×68	4×93	8×57	7×64
9×78	6×95	7×49	8×86

Part 2 · Two digits by two digits**F5 Four boxes instead of two**Both numbers get split, so the grid has **four** boxes.

23×14 , once, in full

×	20	3
10	200	30
4	80	12

$$200 + 30 + 80 + 12 = 322$$

Four boxes, four multiplications, then add all four.

Three boxes means one was missed, and the answer comes out far too small.

×	30	2
10		
5		

(a) 32×15
= _____

×	40	6
20		
3		

(b) 46×23
= _____

×	50	7
30		
4		

(c) 57×34
= _____

×	60	8
40		
2		

(d) 68×42
= _____

×	70	4
20		
6		

(e) 74×26
= _____

×	80	5
30		
7		

(f) 85×37
= _____

F6 Draw the grid yourself

Twelve of them. Draw a four-box grid in each square.

24×13	35×21	47×16
52×34	63×27	78×45
86×53	94×38	75×64

29×37	56×48	83×26
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F7 Use it**Word help – the words that mean multiply**

each · every · per one of them costs this much, so all of them cost this much times the number

altogether · in total the whole lot added up

rows of · boxes of this many groups, each the same size

1. Mr Bowen buys **24** boxes of pencils. Every box holds **36** pencils. How many pencils altogether?

2. A classroom has **18** rows of chairs with **15** chairs in each row. How many chairs are there?

3. A bus has **47** seats. A school books **23** buses for a trip. How many seats are there in total?

4. Mr Bowen keeps **18** fish tanks. Every tank holds **25** rubber ducks. How many rubber ducks does he own?

Part 3 · Dividing – the multiplication backwards

F8 The same three numbers, four ways

Every multiplication you know is also two divisions. The first one is done for you.

Word help – the words that mean divide

share · split between	divide it up so everybody gets the same
goes into	“6 goes into 54 nine times” means $54 \div 6 = 9$
how many each	the answer to a division
left over · remainder	what will not divide up evenly

Multiplication	The other one	Division	The other one
$6 \times 9 = 54$	$9 \times 6 = 54$	$54 \div 6 = 9$	$54 \div 9 = 6$
$7 \times 8 = 56$			
		$72 \div 8 = \underline{\quad}$	
			$96 \div 12 = \underline{\quad}$

F9 Divide

- (a) $84 \div 7 = \underline{\quad}$ (c) $72 \div 6 = \underline{\quad}$ (e) $128 \div 4 = \underline{\quad}$ (g) $192 \div 8 = \underline{\quad}$
 (b) $96 \div 8 = \underline{\quad}$ (d) $135 \div 9 = \underline{\quad}$ (f) $161 \div 7 = \underline{\quad}$ (h) $216 \div 6 = \underline{\quad}$

Now four with something **left over**. Write them like this: $75 \div 8 = 9 \text{ r } 3$.

- (i) $58 \div 7 = \underline{\quad}$ (k) $85 \div 9 = \underline{\quad}$
 (j) $100 \div 6 = \underline{\quad}$ (l) $143 \div 4 = \underline{\quad}$

F10 Find the missing number

Every box holds one number. To find it, **divide**.

- (a) $6 \times \square = 54$ (c) $9 \times \square = 108$ (e) $12 \times \square = 60$
 (b) $\square \times 7 = 84$ (d) $\square \times 8 = 96$ (f) $\square \times 5 = 115$

F11 Share it out

1. **144** pencils are shared equally between **12** students. How many does each student get?

2. **90** students go on a trip. Each bus holds **16** students. How many buses are needed?
Careful: the last bus will not be full, and nobody can be left behind.

3. Mr Bowen shares **216** stickers equally between **8** classes. How many stickers does each class get?

4. Mr Bowen buys **175** rubber ducks and puts the same number into each of his **18** fish tanks. How many go into each tank? How many are left over?

Part 4 • The same grid, with a letter in it

F12 A letter is a number you do not know yet

Collect the like terms. The first one is done for you.

Two rules, and that is all of it

Only the same letter can be added. $3a + 2a = 5a$, but $3a + 2b$ **cannot be added** and is already finished.

A letter on its own means one of it. $5x - x = 5x - 1x = 4x$, **not 5**.

(a) $3a + 2a = 5a$

(c) $7c - 2c = \underline{\hspace{2cm}}$

(e) $m + m = \underline{\hspace{2cm}}$

(b) $4b + 5b = \underline{\hspace{2cm}}$

(d) $5x - x = \underline{\hspace{2cm}}$

(f) $6y - y = \underline{\hspace{2cm}}$

(g) $2p + 3q =$ _____

(h) $8k - 3k =$ _____

(i) $4n + 7 =$ _____

F13 One box has a letter in it

Exactly the grid from Part 1. One box is a letter, so you cannot work that box out — **leave it as it is**.

$4(x + 3)$ is 4×16 with the 16 hidden

$\times$	x	3
4	$4x$	12

$4(x + 3) = 4x + 12$

$4 \times x$ is **$4x$** — you cannot finish it, so you write it and move on. 4×3 is 12. Then **add the boxes**, the same as always.

$\times$	x	5
2		

(a) $2(x + 5)$
 $=$ _____

$\times$	m	4
3		

(b) $3(m + 4)$
 $=$ _____

$\times$	y	2
6		

(c) $6(y + 2)$
 $=$ _____

$\times$	k	7
5		

(d) $5(k + 7)$
 $=$ _____

F14 Fill the boxes

No grid. Write the two numbers straight into the boxes.

(a) $3(a + 2) = \square a + \square$

(d) $2(d + 9) = \square d + \square$

(b) $5(b + 4) = \square b + \square$

(e) $4(e + 6) = \square e + \square$

(c) $7(c + 1) = \square c + \square$

(f) $8(f + 3) = \square f + \square$

F15 A minus inside

The minus sign belongs to the number after it, and it goes into the brackets with that number.

Same grid, one sign

$3(x - 2) = 3x - 6$

$5(y - 4) = 5y - 20$

(a) $4(x - 3) =$ _____

(d) $9(n - 1) =$ _____

(b) $6(m - 2) =$ _____

(e) $3(p - 5) =$ _____

(c) $2(k - 7) =$ _____

(f) $7(q - 2) =$ _____

F16 Find the number outside

This is **F10** again, with a bracket around it. Divide, exactly as you did there.

(a) $\square (x + 2) = 5x + 10$

(d) $\square (k + 5) = 6k + 30$

(b) $\square (y + 3) = 4y + 12$

(e) $\square (p + 4) = 3p + 12$

(c) $\square (m + 1) = 7m + 7$

(f) $\square (n + 6) = 8n + 48$

F17 Check it with a number

This one proves the whole sheet

You wrote $4(x + 3) = 4x + 12$ in F13. Now **put $x = 10$ into both sides** and see whether they agree.

(a) The **left** side. $x + 3$ is $10 + 3 = \square$, so $4(x + 3)$ is $4 \times \square = \square$

(b) The **right** side. $4x$ is $4 \times 10 = \square$, so $4x + 12$ is $\square + 12 = \square$

(c) Did the two sides agree? _____

(d) You have just worked out 4×13 twice, in two different ways. Which two numbers did you add together the second time? _____

(e) Now do the whole thing again on your own, with $3(m + 4)$ and $m = 20$.

(i) Expand it. _____

(ii) The **left** side: work out $m + 4$ first, then multiply by 3. _____

(iii) The **right** side: put $m = 20$ into your expanded answer. _____

(iv) Did they agree? _____

Answer Key – Teacher Copy

F1 Tens and ones

(b) $20 + 3$ (c) $60 + 5$ (d) $10 + 8$ (e) $50 + 2$ (f) $90 + 6$ (g) $30 + 4$ (h) $70 + 0$ (i) $80 + 9$

(h) is the row to check. $70 = 70 + 0$, and a student who cannot write it has been splitting by copying the digits rather than by reading the place value. Accept “70 and nothing”; do not accept 7 and 0.

F2 Times a ten

(a) 120 (b) 300 (c) 280 (d) 540 (e) 160 (f) 270
(g) 600 (h) 2000 (i) 4200 (j) 2400 (k) 2500 (l) 1800

(g) to (l) are the ones that decide whether F5 works, because the top-left box of every two-by-two grid is a ten times a ten and is *much* the biggest number in the grid. A student who writes 60 for (g) will lose a whole place value on every question in Part 2 and their answers will be about ten times too small. That symptom is worth memorising: **answers ten times too small means F2, not F5.**

F3 One row: two digits by one digit

(a) $140 + 21 = 161$ (b) $240 + 48 = 288$
(c) $400 + 32 = 432$ (d) $270 + 63 = 333$

Mark the boxes, not the total. The point of a printed grid is that every step is visible, so a wrong answer here can always be traced to one box.

Both boxes filled and the addition wrong is an arithmetic slip. One box filled is the error this sheet exists to fix.

F4 Now without the printed grid

204 576 595 414
340 372 456 448
702 570 343 688

In order: 6×34 , 8×72 , 7×85 , 9×46 ; 5×68 , 4×93 , 8×57 , 7×64 ; 9×78 , 6×95 , 7×49 , 8×86 . The instruction is to *draw the grid*, and it is worth insisting on for another week even from students who no longer need it – Part 4 asks for the same picture with a letter in it, and a student who has stopped drawing it has nothing to fall back on when the box will not multiply.

F5 Four boxes instead of two

(a) $300 + 20 + 150 + 10 = 480$ (b) $800 + 120 + 120 + 18 = 1058$
(c) $1500 + 210 + 200 + 28 = 1938$ (d) $2400 + 320 + 120 + 16 = 2856$
(e) $1400 + 420 + 80 + 24 = 1924$ (f) $2400 + 560 + 150 + 35 = 3145$

Count the filled boxes before reading the answer. Four numbers means the method was followed; three means one box was missed, which is the single commonest error in two-by-two multiplication and is invisible in the total.

In (b), 23 splits as $20 + 3$ and both rows have to reach both columns. The box that gets forgotten is nearly always the bottom-right one – the small ones times the small ones – because it looks least important and is the easiest to do.

F6 Draw the grid yourself

312 735 752
 1768 1701 3510
 4558 3572 4800
 1073 2688 2158

In order: 24×13 , 35×21 , 47×16 ; 52×34 , 63×27 , 78×45 ; 86×53 , 94×38 , 75×64 ; 29×37 , 56×48 , 83×26 .

$75 \times 64 = 4800$ is a good one to put on the board at the end, because the grid gives $4200 + 300 + 280 + 20$ and the round answer looks like a coincidence until the four boxes are shown.

F7 Use it

1. $24 \times 36 = \mathbf{864}$ pencils. 2. $18 \times 15 = \mathbf{270}$ chairs.
3. $47 \times 23 = \mathbf{1081}$ seats. 4. $18 \times 25 = \mathbf{450}$ rubber ducks.

None of the four says *multiply*. The words doing the work are *every box holds*, *in each row*, *in total* and *every tank holds*, which are all in the word help. A student who adds $24 + 36$ has read the numbers and not the sentence — that is a reading error, and it should be named as one out loud rather than marked as arithmetic.

3 is the hardest arithmetic on the sheet and the grid earns its keep: $800 + 140 + 210 + 21$.

4 is meant to be read completely straight. If the class laughs, that is fine; the question is still 18×25 , and 450 rubber ducks is the correct answer.

F8 The same three numbers, four ways

Row 2: $7 \times 8 = 56$, $8 \times 7 = 56$, $56 \div 7 = 8$, $56 \div 8 = 7$.

Row 3: $8 \times 9 = 72$, $9 \times 8 = 72$, $72 \div 8 = 9$, $72 \div 9 = 8$.

Row 4: $8 \times 12 = 96$, $12 \times 8 = 96$, $96 \div 8 = 12$, $96 \div 12 = 8$.

Rows 3 and 4 are filled in *backwards*, from a division to the multiplication that made it, which is the whole idea Part 3 is for: a division question is a multiplication question you already know the answer to.

This is F10 and F16 in advance. If a student can do this table, neither of those is new work.

F9 Divide

- (a) 12 (b) 12 (c) 12 (d) 15 (e) 32 (f) 23 (g) 24 (h) 36
 (i) 8 r 2 (j) 16 r 4 (k) 9 r 4 (l) 35 r 3

(a), (b) and (c) are all 12 on purpose. Say so afterwards — a student who trusted their arithmetic and got three identical answers will have assumed they were wrong, and that assumption is worth naming.

(f) $161 \div 7$ is the hardest of the exact ones. Counting up in sevens is a perfectly good method and should not be marked down.

F10 Find the missing number

- (a) 9 (b) 12 (c) 12 (d) 12 (e) 5 (f) 23

Every one of these is found by dividing — $54 \div 6$, $84 \div 7$, $108 \div 9$, $96 \div 8$, $60 \div 12$, $115 \div 5$. Say that sentence out loud when you hand the sheet back, because F16 is this question with a bracket drawn round it and the connection is the point.

A student who gets these by guessing and checking has still got them right. Ask for the division as well, once.

F11 Share it out

- $144 \div 12 = 12$ pencils each.
- $90 \div 16 = 5 \text{ r } 10$, so **6 buses** are needed.
- $216 \div 8 = 27$ stickers each. 4. $175 \div 18 = 9$ each, with **13 left over**.

Question 2 is the one worth discussing. The arithmetic gives 5 remainder 10 and the answer is 6, because ten students cannot be left in the car park. Expect 5, and expect “5.6”.

Then put 2 and 4 side by side. Both divisions leave a remainder and the two remainders are treated in opposite ways: the ten students need a sixth bus, and the thirteen spare ducks simply stay in the box. Nothing in the arithmetic tells you which — only the sentence does. That is the whole reason these are word problems and not sums.

4 is meant to be read straight, and 9 r 13 is the full answer; a student who writes 9 has answered only half the question.

F12 A letter is a number you do not know yet

(b) $9b$ (c) $5c$ (d) $4x$ (e) $2m$ (f) $5y$ (g) $2p + 3q$ **finished** (h) $5k$ (i) $4n + 7$ **finished**

(d), (e) and (f) are the invisible 1, three times. (d) comes out 5 if the x is read as nothing; (f) comes out 6 the same way.

(g) and (i) are the two to leave alone, and they are the reason the letter work comes before the brackets: a student who turns $4n + 7$ into $11n$ will turn $4(x + 3)$ into $4x + 12$ and then into $16x$.

F13 One box has a letter in it

(a) $2x + 10$ (b) $3m + 12$ (c) $6y + 12$ (d) $5k + 35$

The left box is the whole lesson. $2 \times x$ cannot be worked out, so it is written as $2x$ and left — and a student who stalls there has not misunderstood multiplication, they have been taught that a box must contain a number. Say plainly that this box does not.

Compare (c) with F3(d) if anyone doubts the grids are the same picture: identical layout, identical instruction, one box that happens to be a letter.

F14 Fill the boxes

(a) $3a + 6$ (b) $5b + 20$ (c) $7c + 7$ (d) $2d + 18$ (e) $4e + 24$ (f) $8f + 24$

(c) is the one that looks wrong and is not: $7 \times 1 = 7$, so both boxes hold a 7. Expect it to be left blank. The printed boxes are doing real work here. A blank square in front of the letter is much harder to skip than an empty line, and skipping the second number — $3(a + 2) = 3a + 2$ — is the headline error of Maths 2.4.

F15 A minus inside

(a) $4x - 12$ (b) $6m - 12$ (c) $2k - 14$ (d) $9n - 9$ (e) $3p - 15$ (f) $7q - 14$

(d) is the invisible 1 arriving inside a bracket: $9 \times 1 = 9$, so the answer is $9n - 9$ and not $9n - 1$. It is F14(c) with a minus sign, and a student who got that one right and this one wrong was copying rather than multiplying.

F16 Find the number outside

(a) 5 (b) 4 (c) 7 (d) 6 (e) 3 (f) 8

Found by dividing: $10 \div 2$, $12 \div 3$, $7 \div 1$, $30 \div 5$, $12 \div 4$, $48 \div 6$. Or by dividing the other term — $5x \div x$ — which is also right and is worth hearing if a student offers it.

Point back at F10 before they start. The two questions are the same question and the sheet says so; a student who does F10 confidently and stalls here has been stopped by the bracket, not by the arithmetic.

F17 Check it with a number

(a) $10 + 3 = 13$, so $4 \times 13 = 52$.

(b) $4 \times 10 = 40$, so $40 + 12 = 52$.

(c) **Yes** — both sides are 52.

(d) **40 and 12.**

(e) (i) $3m + 12$ (ii) $m + 4 = 24$, and $3 \times 24 = 72$ (iii) $3 \times 20 = 60$, and $60 + 12 = 72$ (iv) **yes.**

(e) is (a) to (d) with the scaffolding taken away, and it is the one that shows whether the method transferred. The two boxes this time are **60 and 12**, which is the grid for 3×24 .

This is the sheet in one question. The left side is 4×13 done as a single multiplication. The right side is 4×13 done as $40 + 12$ — which is the grid from F3, with the same two boxes, arrived at from the algebra.

(d) is the part to ask for out loud. A student who says “40 and 12” and then sees that those are the two boxes of 4×13 has understood why expanding a bracket is not a new rule but the multiplication they have done all year, written down before the letter is known.

If there is time for one thing from this sheet at the board, do this question and draw the grid for 4×13 beside it.